

$$1 \text{ a) } R = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$b) \dim N(A) = 2$$

$$\dim C(A) = 2$$

$$\dim C(A^T) = 2$$

$$\dim N(A^T) = 1$$

$$N(A) = c_1 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

$$C(A^T) = c_1 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

2 a) $U^{-1} = \begin{bmatrix} +a & -b & -c \\ 0 & +d & -e \\ 0 & 0 & +f \end{bmatrix}$ X

b) $A = U \Lambda U^{-1}$ Product of upper triangulars. ✓

(c) ~~columns are orthogonal~~ ~~columns are orthogonal~~

~~columns are not orthogonal~~ ~~columns are not orthogonal~~

The columns are not orthogonal. ✓

3 a) $r(\text{Block}) = \max_{\Lambda} \text{rank}(A_{\Lambda}/B)$ X

b) $r(A) = r(B)$ Squaring Preserves rank. ✓

c) $n-r$, ~~rank~~ $2(n-r)$ ✓ X

4 a) Indep ✓

b) $x^T A^T A x$ is $|Ax|^2$ ✓

c) left, $BA = (A^T A)^{-1} A^T A = I$ ✓

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5 a) $x^T x = c_1^2 + c_2^2 + c_3^2$ ✓

$x^T A x = \lambda_1 c_1^2 + \lambda_2 c_2^2 + \lambda_3 c_3^2$ ✓

b) q_h ✓

6 a) $0 = w$ ✓

b) $q_i = \frac{u_i}{|u|}$ $r_i = \begin{bmatrix} |u_i| \\ \sigma \end{bmatrix}$ ✓

c) $Q R (R^T Q^T Q R)^{-1} (R^T Q^T) = Q R R^T Q^T = Q Q^T$ ✓

7 a) $1, -1, i, -i$ ✓

b) c^T, c^{2T} ✓

c) $1x - 1 \times i x - i = -1$ ✓

8 a) $A^T A \hat{x} = A^T b$ ✓

$\hat{x} = \frac{A^T b}{A^T A}$ ✓

$P = A \frac{A^T b}{A^T A}$ ✓

$A A^T (A^T A)^{-1} b$

b) $C(A)$ ✓

$N(A^T)$ ✓

c) $A^* (A^T A)^{-1} A^T$ ✓

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Problem 9

a) $7, 3 \times 7 - 2 \times 3 = 15 \checkmark$

b) $D_n = 3 D_{n-1} - 2 D_{n-2} \checkmark$

c) $\begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix} \lambda = 1, 2$
by a factor of $2^n \checkmark$